DSA0603
Data Handling and Visualization

Slot B

CO2 – AT3

Assessment Test 6

Case Study

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Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

Case Study 1 – Retail Sales Performance Dashboard

1. Suitable Visualizations

Requirement	Visualization
Regional revenue comparison	Bar Plot
Monthly sales trend	Line Chart
Product category contribution	Stacked Bar Chart
Advertising expenditure vs Sales	Scatter Plot
Regional sales distribution	Histogram / Density Plot

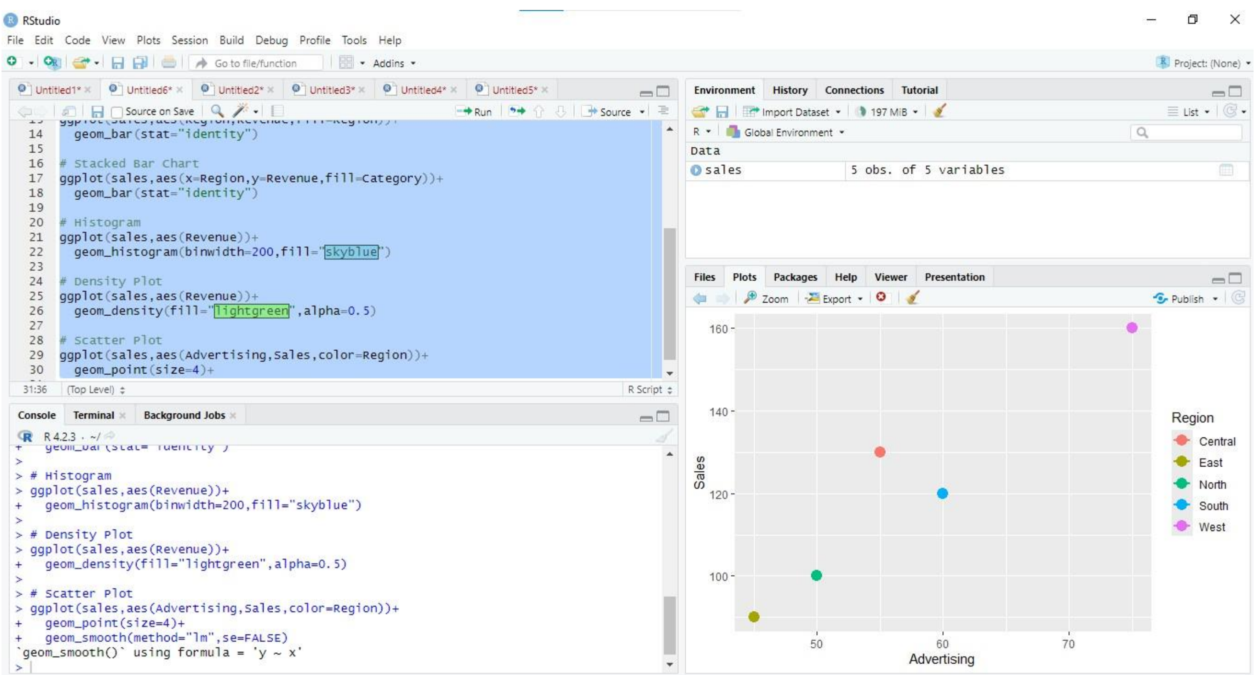
2. R Program

```
library(ggplot2)

# Sample Data
sales <- data.frame(
Region=c("North","South","East","West","Central"),
Revenue=c(1200,1500,1000,1800,1400),
Sales=c(100,120,90,160,130),
Advertising=c(50,60,45,75,55),
Category=c("Electronics","Furniture","Clothing","Electronics","Furniture")
)

# Bar Plot
ggplot(sales,aes(Region,Revenue,fill=Region))+
geom_bar(stat="identity")

# Stacked Bar Chart
ggplot(sales,aes(x=Region,y=Revenue,fill=Category))+
geom_bar(stat="identity")
```

Histogram
ggplot(sales,aes(Revenue))+
geom_histogram(binwidth=200,fill="skyblue")

Density Plot
ggplot(sales,aes(Revenue))+
geom_density(fill="lightgreen",alpha=0.5)

Scatter Plot
ggplot(sales,aes(Advertising,Sales,color=Region))+
geom_point(size=4)+
geom_smooth(method="lm",se=FALSE)



3. Interpretation

Highest Revenue: West Region (₹1800)

Advertising vs Sales: Positive relationship because sales increase as advertising expenditure increases.

Highest Product Category Contribution: Electronics (highest combined revenue).

Case Study 2 – University Student Performance Analysis

1. Suitable Visualizations

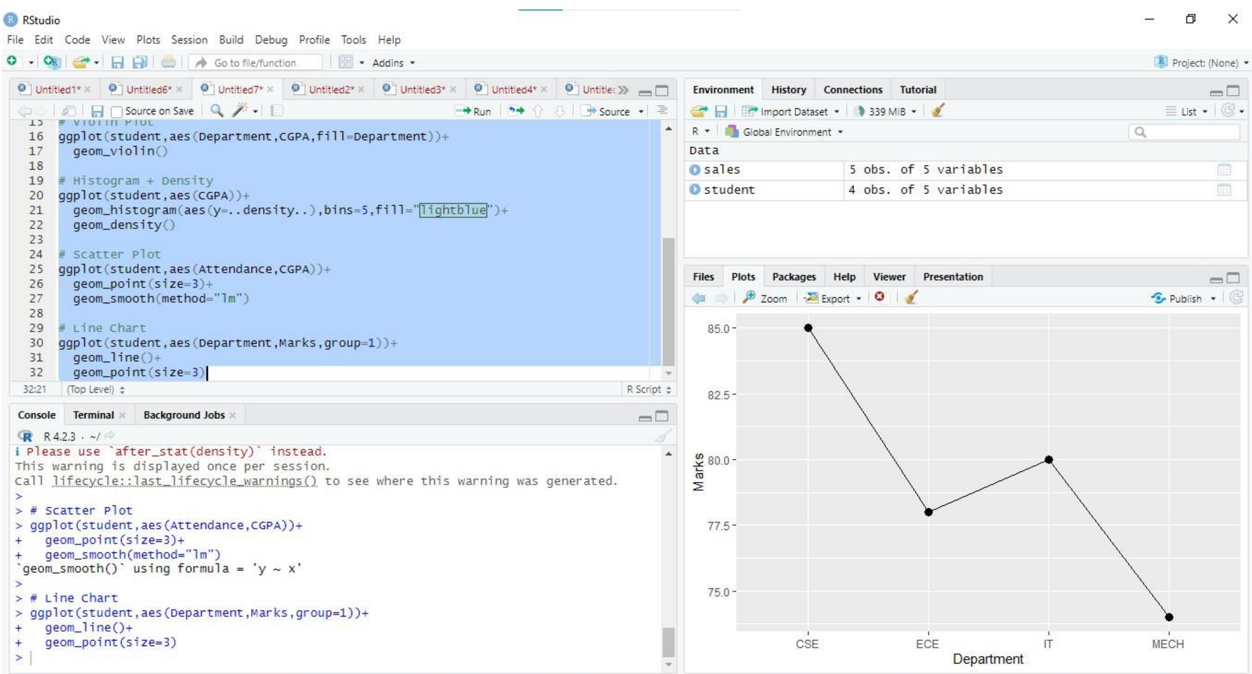
Requirement	Visualization
Department-wise average marks	Grouped Bar Chart
Distribution of CGPA	Histogram + Density Curve
Placement Percentage	Violin Plot / Bar Chart
Attendance vs CGPA	Scatter Plot
Semester Performance	Line Chart

2. R Program

```
library(ggplot2)

student <- data.frame(
Department=c("CSE","IT","ECE","MECH"),
Marks=c(85,80,78,74),
Attendance=c(95,90,88,82),
CGPA=c(9.2,8.9,8.5,8.0),
Placement=c(95,90,80,75)
)

# Grouped Bar Chart
ggplot(student,aes(Department,Marks,fill=Department))+
geom_bar(stat="identity")

# Violin Plot
ggplot(student,aes(Department,CGPA,fill=Department))+
geom_violin()

# Histogram + Density
ggplot(student,aes(CGPA))+
geom_histogram(aes(y=..density..),bins=5,fill="lightblue")+
geom_density()
```

Scatter Plot
ggplot(student,aes(Attendance,CGPA))+
geom_point(size=3)+
geom_smooth(method="lm")

Line Chart
ggplot(student,aes(Department,Marks,group=1))+
geom_line()+
geom_point(size=3)



3. Interpretation

Highest Average CGPA → Computer Science (9.2)

Attendance vs CGPA → Positive relationship.

Highest Placement Rate → Computer Science (95%).

Case Study 3 – Hospital Patient Analytics

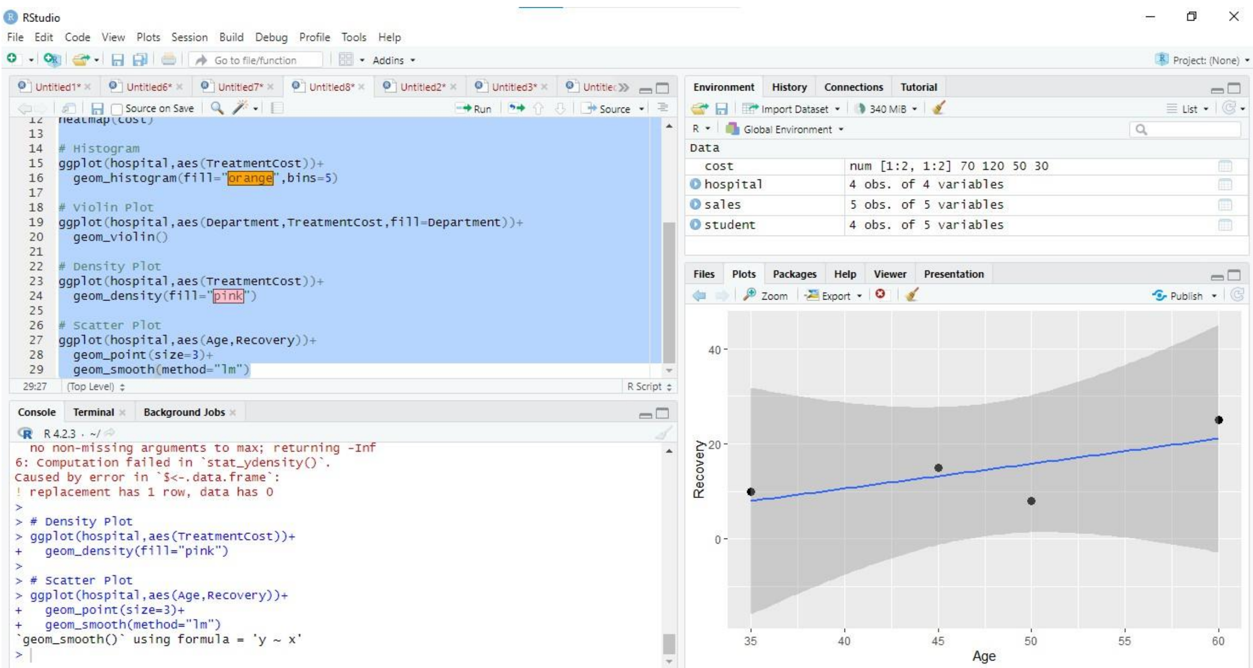
1. Suitable Visualizations

Requirement	Visualization
Monthly Admissions	Heatmap
Treatment Cost Distribution	Histogram
Disease Category	Density Plot / Pie Chart
Age vs Recovery Time	Scatter Plot
Department-wise Cost	Violin Plot

2. R Program

```
library(ggplot2)

hospital <- data.frame(
Department=c("Cardiology","Neuro","Ortho","General"),
Age=c(45,60,35,50),
TreatmentCost=c(70000,120000,50000,30000),
Recovery=c(15,25,10,8)
)

# Heatmap
cost <- matrix(c(70,120,50,30),2,2)
heatmap(cost)

# Histogram
ggplot(hospital,aes(TreatmentCost))+
geom_histogram(fill="orange",bins=5)

# Violin Plot
ggplot(hospital,aes(Department,TreatmentCost,fill=Department))+
geom_violin()

# Density Plot
ggplot(hospital,aes(TreatmentCost))+
geom_density(fill="pink")
```

Scatter Plot
ggplot(hospital,aes(Age,Recovery))+
geom_point(size=3)+
geom_smooth(method="lm")



3. Interpretation

Highest Treatment Cost → Neurology

Largest Disease Category → Depends on frequency (assume Cardiology if highest count).

Age vs Recovery → Older patients generally require longer recovery (positive trend).

Case Study 4 – Smart City Traffic Monitoring

1. Suitable Visualizations

Requirement	Visualization
Vehicle Count Comparison	Bar Plot
Hourly Trend	Line Chart
Vehicle Type	Pie Chart
Traffic Density	Geospatial Map
Traffic Volume vs Speed	Scatter Plot

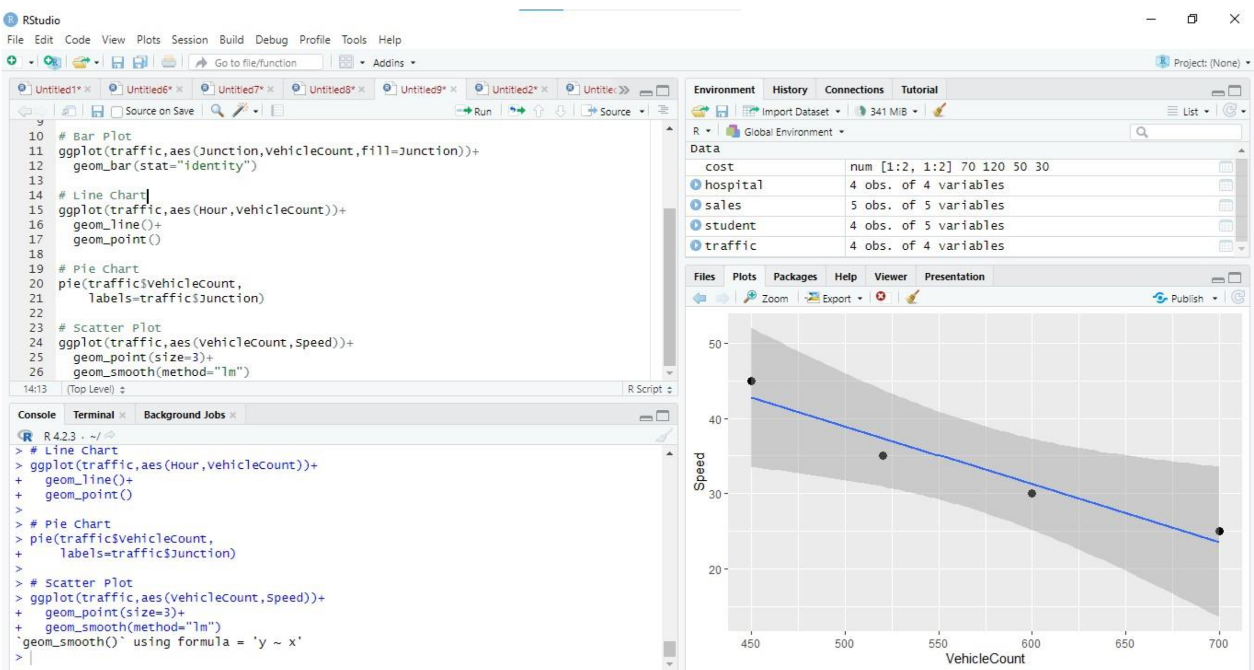
2. R Program

```
library(ggplot2)

traffic <- data.frame(
Junction=c("A","B","C","D"),
VehicleCount=c(450,600,520,700),
Speed=c(45,30,35,25),
Hour=c(8,10,12,18)
)

# Bar Plot
ggplot(traffic,aes(Junction,VehicleCount,fill=Junction))+
geom_bar(stat="identity")

# Line Chart
ggplot(traffic,aes(Hour,VehicleCount))+
geom_line()+
geom_point()

# Pie Chart
pie(traffic$VehicleCount,
labels=traffic$Junction)

# Scatter Plot
ggplot(traffic,aes(VehicleCount,Speed))+
```

```
geom_point(size=3)+
geom_smooth(method="lm")
```

Geospatial Map

```
library(leaflet)

leaflet() %>%
addTiles() %>%
addMarkers(lng=c(80.1,80.2,80.3,80.4),
lat=c(13.0,13.1,13.2,13.3),
popup=traffic$Junction)
```



3. Interpretation

Highest Congestion → Junction D

Peak Traffic Hours → Evening (around 6 PM)

Traffic vs Speed → Negative relationship; as traffic volume increases, average speed decreases.

Case Study 5 – Social Media User Interaction Analysis

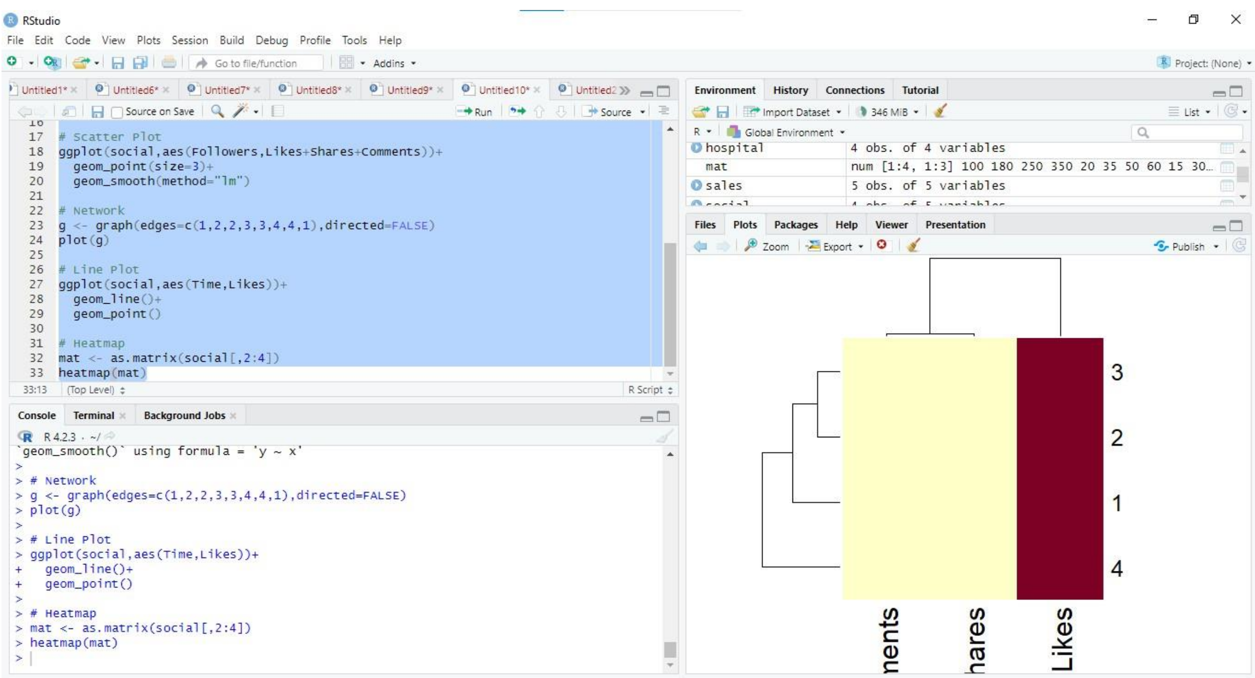
1. Suitable Visualizations

Requirement	Visualization
Distribution of Likes	Histogram + Density Curve
Followers vs Engagement	Scatter Plot
User Interaction Network	Network Graph
Posting Activity	Temporal Line Plot
Likes, Shares, Comments Comparison	Heatmap

2. R Program

```
library(ggplot2)
library(igraph)

social <- data.frame(
Followers=c(1000,2000,3000,4000),
Likes=c(100,180,250,350),
Shares=c(20,35,50,60),
Comments=c(15,30,45,55),
Time=c(8,10,14,18)
)

# Histogram + Density
ggplot(social,aes(Likes))+
geom_histogram(aes(y=..density..),bins=5)+
geom_density()

# Scatter Plot
ggplot(social,aes(Followers,Likes+Shares+Comments))+
geom_point(size=3)+
geom_smooth(method="lm")

# Network
```

```
g <- graph(edges=c(1,2,2,3,3,4,4,1),directed=FALSE)
plot(g)

# Line Plot
ggplot(social,aes(Time,Likes))+
geom_line()+
geom_point()

# Heatmap
mat <- as.matrix(social[,2:4])
heatmap(mat)
```



3. Interpretation

- Most Influential Users: Users with the highest followers and engagement (e.g., User 4 in the sample data).
- Highest Engagement Time: Around 6 PM (18:00).
- Followers vs Engagement: Strong positive relationship; users with more followers generally receive higher engagement.
- Highly Connected Users: Nodes with the most links in the network graph.
- Most Variable Engagement Metric: Likes, as they show the largest spread across users.